

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Examination – Winter 2024**

**Course: B. Tech. Branch: Mechanical Engineering/ Mechanical Engineering(Sandwich)**

**Subject Code & Name: BTMEC704B-Industrial Engineering & Management Semester: VII**

**Max Marks: 60**

**Date:12/02/2025**

**Duration: 3 Hr**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 is compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

|                                                                                                                                                                         |     |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|
| <b>Q. 1 Objective type questions. (Compulsory Question)</b>                                                                                                             |     | <b>12</b> |
| 1 Who of the following is related to the contribution in scientific management?                                                                                         | CO1 | 1         |
| a. Bill Gates      b. Henri Fayol      c. Graham Bell      d. Louis Pasteur                                                                                             |     |           |
| 2 Planning is a process by which _____                                                                                                                                  | CO1 | 1         |
| a. manager select, train, promote their staff                                                                                                                           |     |           |
| b. a manager anticipates the future                                                                                                                                     |     |           |
| c. manager achieve harmony of individual efforts                                                                                                                        |     |           |
| d. manager monitor program activities                                                                                                                                   |     |           |
| 3 Staffing is a process by which _____                                                                                                                                  | CO2 | 1         |
| a. a manager anticipates the future                                                                                                                                     |     |           |
| b. manager select, train, promote their staff                                                                                                                           |     |           |
| c. manager achieve harmony of individual efforts                                                                                                                        |     |           |
| d. manager monitor program activities                                                                                                                                   |     |           |
| 4 _____ is the quality of the behavior of the managers whereby they inspire confidence and trust in their subordinates and guide their activities in organized efforts. | CO2 | 1         |
| a. leadership      b. planning      c. staffing      d. forecasting                                                                                                     |     |           |
| 5 Which of the following is not a sales forecasting technique?                                                                                                          | CO3 | 1         |
| a. Market Survey      b. Delphi Method      c. Correlation Analysis      d. Batch Production                                                                            |     |           |
| 6 Which of the following term is not related to Material Requirements Planning (MRP)                                                                                    | CO3 | 1         |
| a. Economic order quantity      b. Bill of Materials file      c. Inventory record file      d. Matching people to jobs                                                 |     |           |
| 7 The facilities layout in which similar machines or similar operations are kept at one location is called as _____                                                     | CO4 | 1         |
| a. Product Layout      b. Combination Layout      c. Process Layout      d. None of these                                                                               |     |           |
| 8 The layout in which the men and equipment are moved to the                                                                                                            | CO4 | 1         |

|             |                                                                                                                                                            |                                   |                                           |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|
|             | material, which remains at one place is called as,                                                                                                         |                                   |                                           |
|             | a. Product Layout                                                                                                                                          | b. Fixed position layout          | c. Process Layout                         |
|             |                                                                                                                                                            |                                   | d. None of these                          |
| 9           | By clearly stating the goals of the business and meshing them with the goals of the employees, managers make use of a management technique called as ..... | CO4                               | 1                                         |
|             | a. Total Quality Management (TQM)                                                                                                                          | b. Management by Objectives (MBO) | c. Total Preventive Maintenance (TPM)     |
|             |                                                                                                                                                            |                                   | d. All of these                           |
| 10          | _____ is concerned with investigating, reducing and eliminating ineffective time, whatever may be the cause                                                | CO5                               | 1                                         |
|             | a. Product Layout                                                                                                                                          | b. Work measurement               | c. Bill of Materials file                 |
|             |                                                                                                                                                            |                                   | d. Economic order quantity                |
| 11          | Motion studies are used to _____                                                                                                                           | CO5                               | 1                                         |
|             | a. Develop the best work method                                                                                                                            | b. Develop motion consciousness   | c. Develop economical and efficient tools |
|             |                                                                                                                                                            |                                   | d. All of these                           |
| 12          | Which of the following is not a Total Quality Management technique?                                                                                        | CO5                               | 1                                         |
|             | a. Just in time (JIT)                                                                                                                                      | b. quality circles                | c. six sigma                              |
|             |                                                                                                                                                            |                                   | d. Bill of Materials                      |
| <b>Q.2</b>  | <b>Solve the following.</b>                                                                                                                                |                                   | <b>12</b>                                 |
| A)          | What are the different functions of Management?                                                                                                            | CO1                               | 6                                         |
| B)          | Explain the concept of decentralization in the organization.                                                                                               | CO1                               | 6                                         |
| <b>Q.3</b>  | <b>Solve the following.</b>                                                                                                                                |                                   | <b>12</b>                                 |
| A)          | Explain the importance of performance appraisal and career strategy.                                                                                       | CO2                               | 6                                         |
| B)          | Briefly explain how team building, and communication help managers in effective leadership.                                                                | CO2                               | 6                                         |
| <b>Q. 4</b> | <b>Solve Any Two of the following.</b>                                                                                                                     |                                   | <b>12</b>                                 |
| A)          | Explain any one type of facility layout in detail.                                                                                                         | CO3                               | 6                                         |
| B)          | What are the different characteristics of services systems?                                                                                                | CO3                               | 6                                         |
| C)          | What is materials requirement planning?                                                                                                                    | CO3                               | 6                                         |
| <b>Q.5</b>  | <b>Write Short Notes on Any Two of the following.</b>                                                                                                      |                                   | <b>12</b>                                 |
| A)          | Capacity planning                                                                                                                                          | CO4                               | 6                                         |
| B)          | Facilities layout                                                                                                                                          | CO4                               | 6                                         |
| C)          | Assembly line balancing                                                                                                                                    | CO4                               | 6                                         |
| <b>Q.6</b>  | <b>Write Short Notes on Any Two of the following.</b>                                                                                                      |                                   | <b>12</b>                                 |
| A)          | Micro motion study                                                                                                                                         | CO5                               | 6                                         |
| B)          | Concurrent engineering                                                                                                                                     | CO5                               | 6                                         |
| C)          | Quality circles                                                                                                                                            | CO5                               | 6                                         |

\*\*\* End \*\*\*